INCH-POUND

MIL-PRF-1/877D
21 September 1999
SUPERSEDING
MIL-E-1/877C
9 May 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 100TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode.

See figure 1.

Weight: 4-ounces (113.4 grams) nominal.

Mounting position: Vertical.

ABSOLUTE RATINGS: C Telegraphy

| Parameter:<br>Unit: | F1<br>MHz | Ef<br>V | Eb<br>V dc | Ec<br>V dc | Ib<br>mA dc | Ic<br>mA dc | Pp<br>W |
|---|---|---|---|---|---|---|---|
| Maximum:<br>Minimum: | 40<br>--- | 5.25<br>4.75 | 3,000<br>--- | -210<br>--- | 225<br>--- | 50<br>--- | 100<br>--- |
| Test conditions: | --- | 5.0 V ac | 2,500 | Adjust | 40 | --- | --- |

See footnotes at end of table I.

GENERAL:

Qualification: Required.


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TABLE I. Testing and inspection.

| Inspection | Method | Notes | Conditions | Acceptance Level 3/ | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | | | |
| Filament current | 1301 | --- | | 0.65 | If | 5.8 | 6.6 | A ac |
| Electrode voltage (1) (grid) | 1261 | --- | | 0.65 | Ec(1) | -34.0 | -46.0 | V dc |
| Electrode voltage (2) (grid) | 1261 | --- | Eb = 14.0 kV dc; Ec/Ib = 1.0 mA dc | 0.65 | Ec(2) | --- | -550 | V dc |
| Total grid current | 1266 | 1/ | | 0.65 | Ic | --- | -10 | µA dc |
| Primary grid emission | 1266 | --- | Ic = 100 mA dc; t = 15 seconds; Ef = 5.5 V ac | 0.65 | Isg | --- | -500 | µA dc |
| Peak emission | 1231 | --- | eb = ec = 2,500 v | 0.65 | is | 3.0 | --- | a |
| Conformance inspection, part 2 | | | | | | | | |
| Amplification factor | 1316 | --- | | --- | Mu | 34.0 | 42.0 | --- |
| Power oscillation | 1236 | --- | F = 8 MHz; Eb = 3,000 V dc; Ib = 150 mA dc | --- | Po | 270 | --- | W (useful) |
| Direct-interelectrode capacitance | 1331 | --- | | --- | Cgp | 1.7 | 2.3 | pF |
| | | | | | Cin | 2.5 | 3.4 | pF |
| | | | | | Cout | --- | 0.45 | pF |
| Conformance inspection, part 3 | | | | | | | | |
| Low-frequency vibration | 1031 | 2/ | No voltages applied | --- | --- | --- | --- | --- |
| Bump | 1036 | 2/ | Angle = 10° | --- | --- | --- | --- | --- |
| Life test | --- | --- | Group C; power oscillation; t = 500 hours | --- | --- | --- | --- | --- |
| Life-test end points: | --- | --- | | | | | | |
| Peak emission | 1231 | --- | | --- | is | 2.5 | --- | a |
| Primary grid emission | 1266 | --- | | --- | Isg | --- | -500 | µA dc |

1/ This test is to be the first test performed at the conclusion of the holding period.

2/ This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of zero. In the event of failure, the test will be made as a part of conformance inspection, part 2, code level D, with an acceptance level of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

3/ This specification uses accept on zero ( c = 0) sampling plan in accordance with MIL-PRF-1, Table III.



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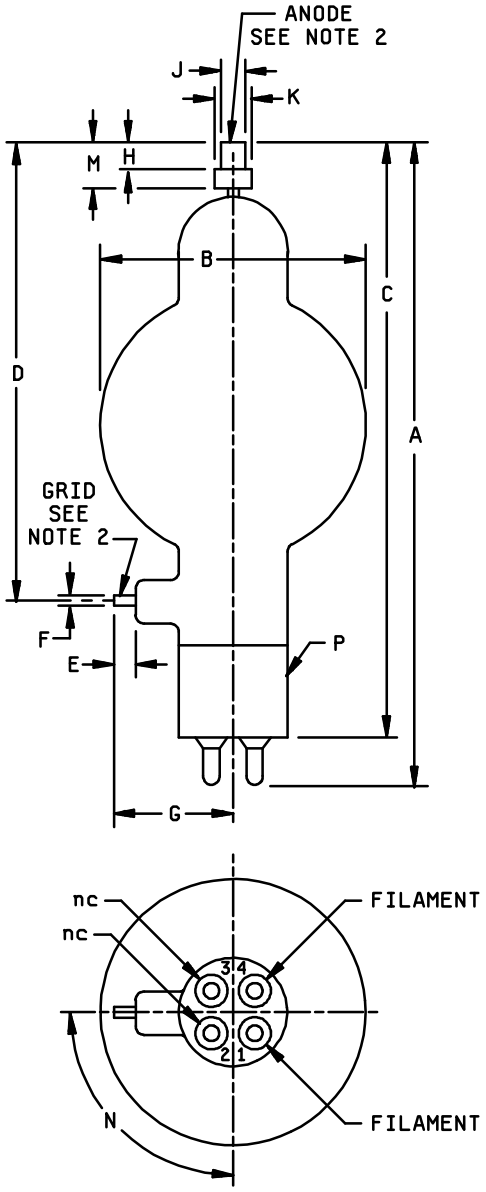


| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | 7.250 | 7.750 | 184.15 | 196.85 |
| B | --- | 3.188 | --- | 80.98 |
| D | 5.000 | 5.500 | 127.00 | 139.70 |
| E | .250 | --- | 6.35 | --- |
| G | 1.250 | 1.500 | 31.75 | 38.10 |
| N | 80° | | 100° | |
| Conformance inspection, part 3 (see note 1) | | | | |
| C | 6.625 | 7.125 | 168.28 | 180.98 |
| F | .055 | .061 | 1.40 | 1.55 |
| J | .350 | .365 | 8.89 | 9.27 |
| P | Base: A4-10 (EIA) | | | |
| Reference dimensions | | | | |
| H | .328 | | 8.33 | |
| KD | .500 | | 12.70 | |
| M | .563 | | 14.30 | |

NOTES:

1. These dimensions shall be checked annually with the following sampling plan:

   n1 = 4 c1 = 0 where c2 represents the total failures for the first
   n2 = 4 c2 = 0; and second samples combined.

   In case of failure after double sampling, the failing dimension(s) shall become conformance inspection, part 2, for three successful consecutive submissions, at which time the test may revert to the conformance inspection, part 3 basis.

2. Heat radiating connectors HR-2 (grid) and HR-6 (anode) or equivalent, are recommended for all applications.

FIGURE 1. Outline drawing of electron tube type 100TH.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

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